
Lecture 4

Digital Image Enhancement

1. Principle of image enhancement
2. Spatial domain transformation
 - Basic intensity transformation
 - Histogram processing

Principle Objective of Enhancement

- Image enhancement is to process an image so that the result will be more suitable than the original image for a specific application.

- Input an image, output another image

$$f(x, y) \xrightarrow{T} g(x, y)$$

$$g = T(f)$$

- The suitability is up to each application.
- A method which is quite useful for enhancing an image may not necessarily be the best approach for enhancing another images

Good images

- For human visual
 - The visual evaluation of image quality is a highly subjective process.
 - It is hard to standardize the definition of a good image.
- For machine perception
 - The evaluation task is easier.
 - A good image is one which gives the best machine recognition results.
- A certain amount of trial and error usually is required before a particular image enhancement approach is selected.

Two Domains

- Spatial Domain
 - Direct manipulation of pixels in an image

$$g(x,y) = T[f(x,y)]$$

- Frequency Domain :



$$T(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) r(x, y, u, v)$$

$$f(x, y) = \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} T(u, v) s(x, y, u, v)$$

- Combinations of spatial domain and frequency domain.

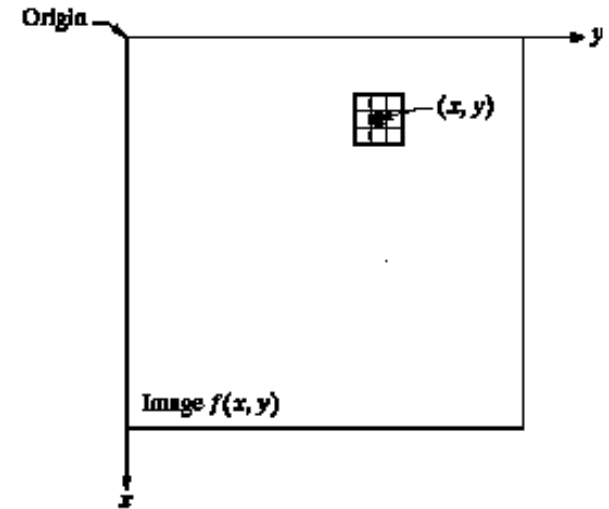
Spatial Domain

- Procedures that operate directly on pixels.

$$g(x,y) = T[f(x,y)]$$

where

- $f(x,y)$ is the input image
- $g(x,y)$ is the processed image
- T is an operator on f defined over some neighborhood of (x, y)



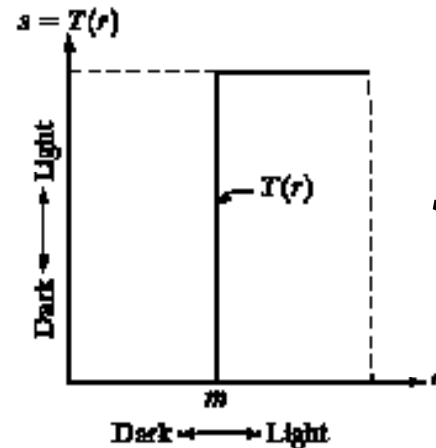
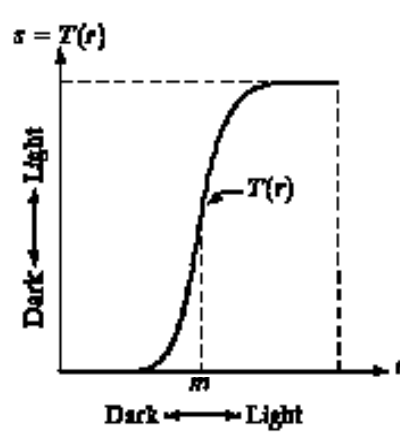
- Neighborhood
 - Given a distance measure D , and a real number r . The r -neighborhood of a point $p_0(x_0, y_0)$ is defined to be the set of pixels

$$N_D^r(p_0) = \{p(x, y) : D(p_0, p) \leq r\}$$

Two classes depending on the neighborhood

- Point Processing (one pixel neighborhood)
 - g only depends on the value of f at (x,y)
 - T is gray level (or intensity or mapping) transformation function
 $s = T(r)$
 r is the gray level of $f(x, y)$, s is the gray level of $g(x, y)$
- Mask/Filter/convolution processing
 - When $r \geq 1$, the r -neighborhood forms a mask or filter.
 - Use a function of the values of f in a predefined neighborhood of (x, y) to determine the value of g at (x, y)
 - The value of the mask coefficients determine the nature of the process
 - Computed by convolution used in techniques for image Sharpening and image Smoothing

Example: Contrast Stretching



$$s = T(r) = \frac{1}{1 + (m/r)^E}$$

- m is the threshold value, E determined the slope.
- Produce higher contrast than the original by
 - darkening the levels below m in the original image
 - Brightening the levels above m in the original image

Three basic gray-level transformation functions

- Linear function
 - Negative and identity transformations
- Logarithm function
 - Log and inverse-log transformation
- Power-law function
 - n^{th} power and n^{th} root transformations

Linear function

- $s = ar + b + c$
- Identity function

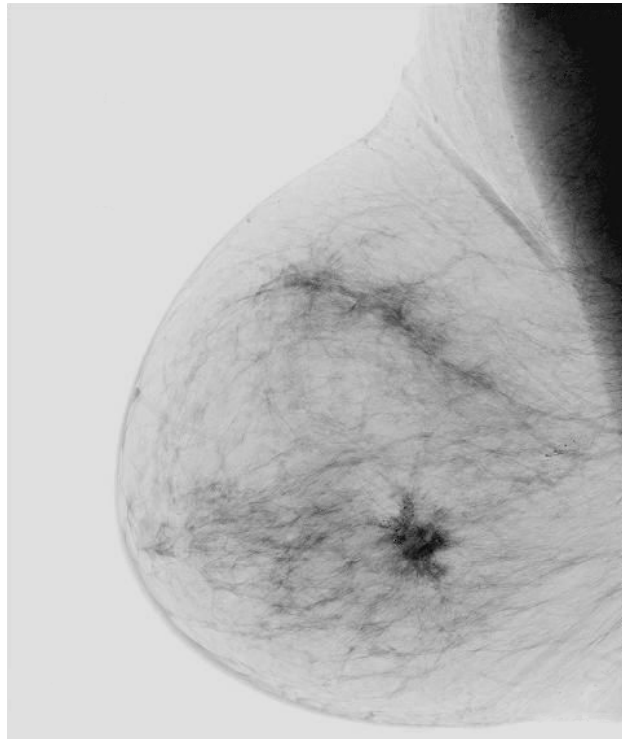
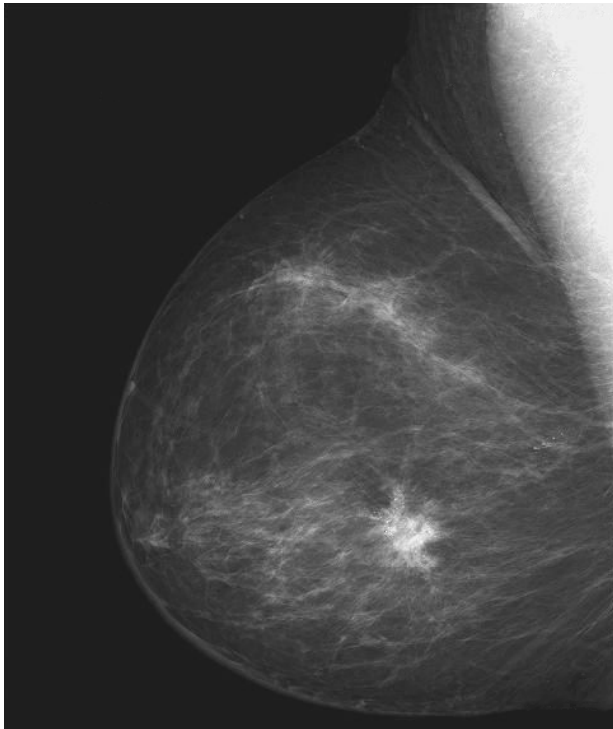
$$s = r$$

- Negative image

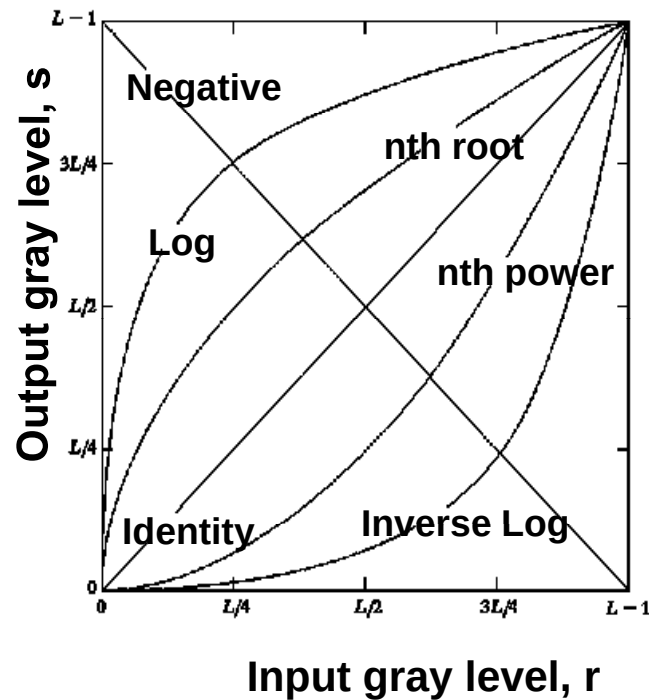
$$s = L - 1 - r$$

An image with gray level in the range $[0, L-1]$
where $L = 2^n$; $n = 1, 2, \dots$

Negative image



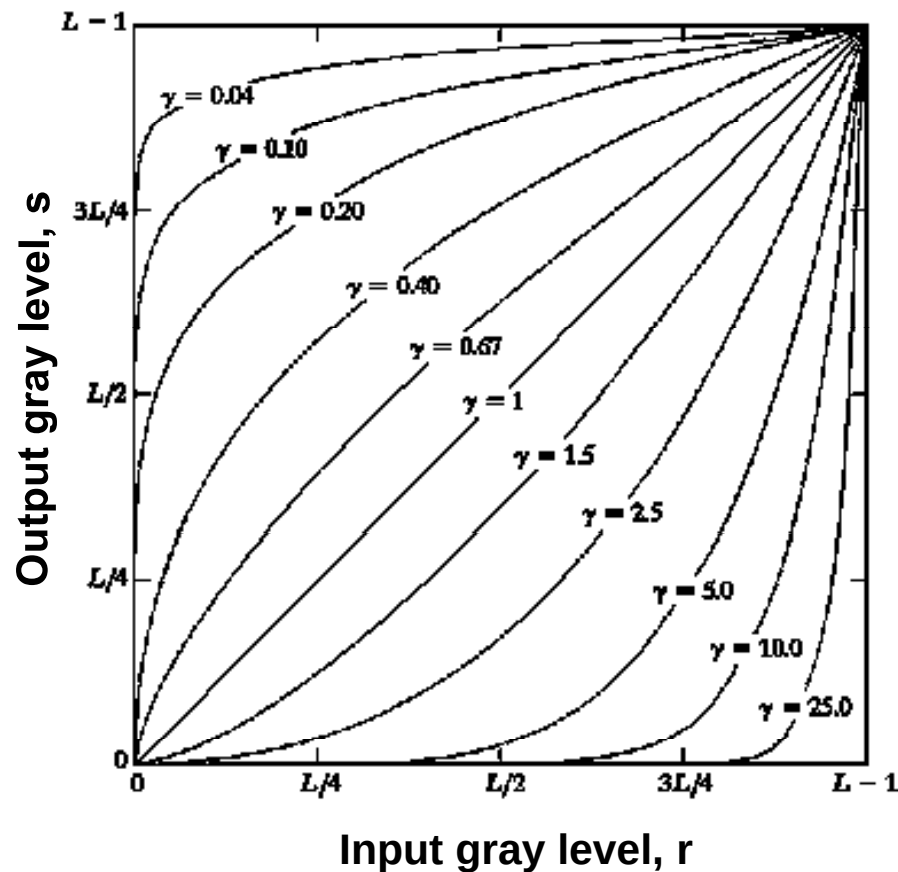
Log Transformations



$$s = c \log (1+r)$$

- c is a constant and $r \geq 0$
- Log curve maps a narrow range of low gray-level values in the input image into a wider range of output levels.
- Used to expand the values of dark pixels in an image while compressing the higher-level values.

Power-Law Transformations

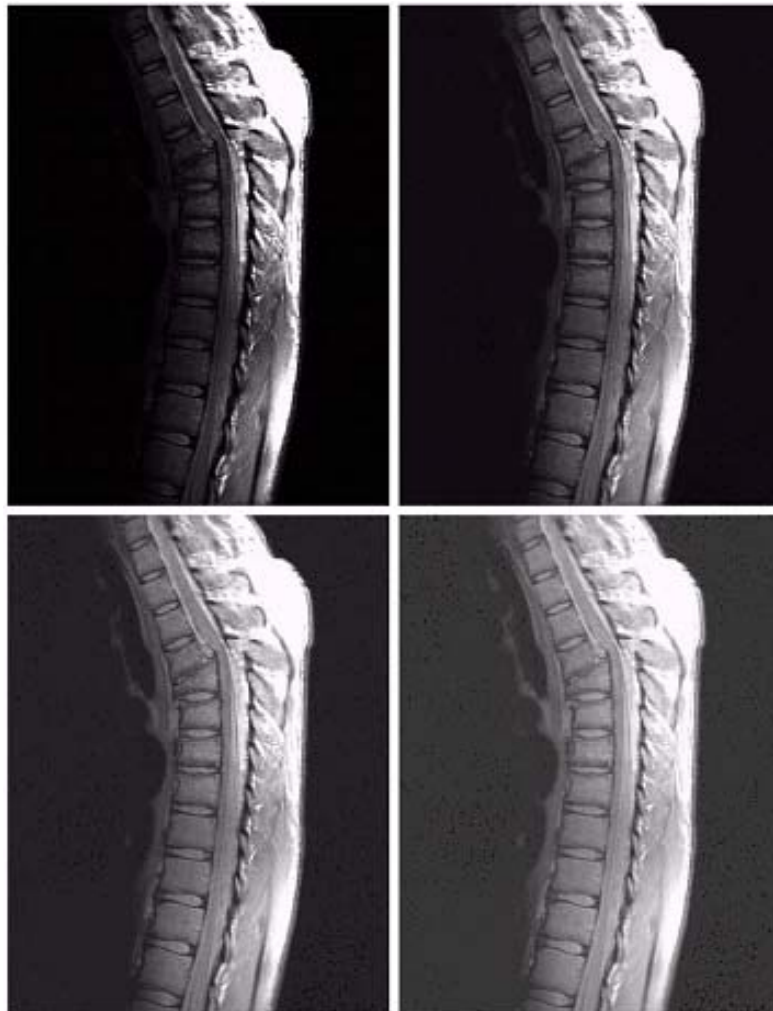


$$s = cr^\gamma$$

- c and γ are positive constants
- Power-law curves with fractional values of γ map a narrow range of dark input values into a wider range of output values, with the opposite being true for higher values of input levels.
- $c = \gamma = 1 \Rightarrow$ Identity function

Another example : MRI

a	b
c	d



(a) a magnetic resonance image of an upper thoracic human spine with a fracture dislocation and spinal cord impingement

- The picture is predominately dark
- An expansion of gray levels are desirable $\Rightarrow$ needs $\gamma < 1$

(b) result after power-law transformation with $\gamma = 0.6$, $c=1$

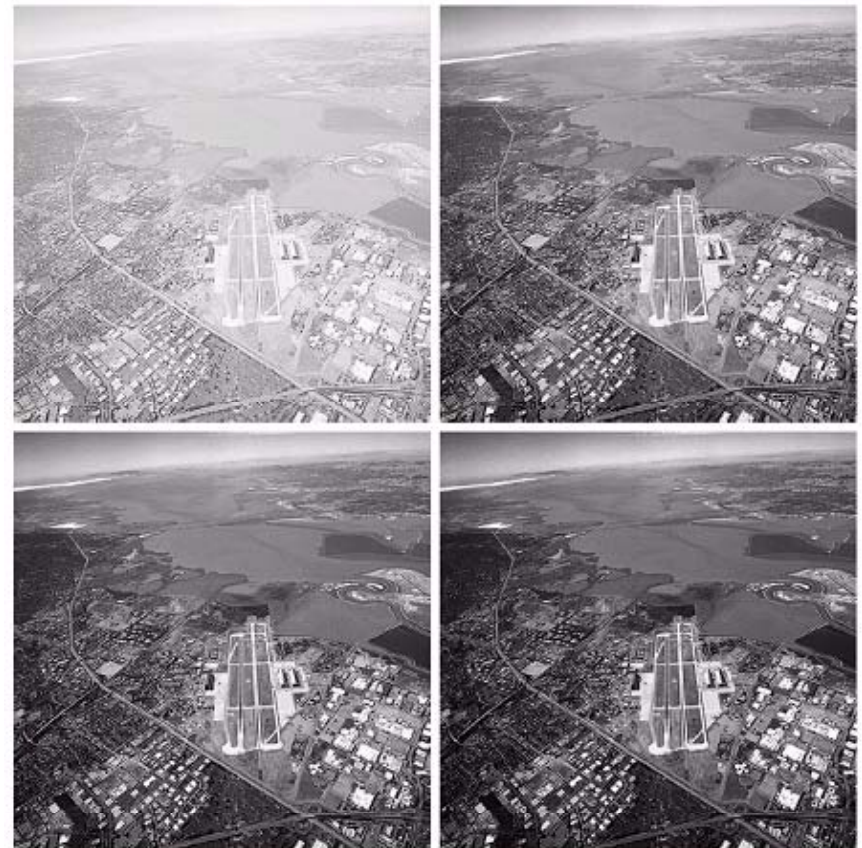
(c) transformation with $\gamma = 0.4$
(best result)

(d) transformation with $\gamma = 0.3$
(under acceptable level)

Another example

a	b
c	d

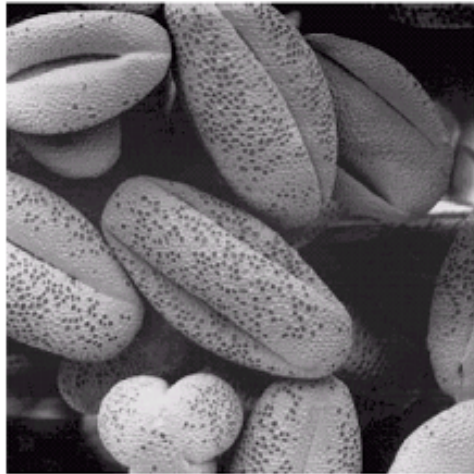
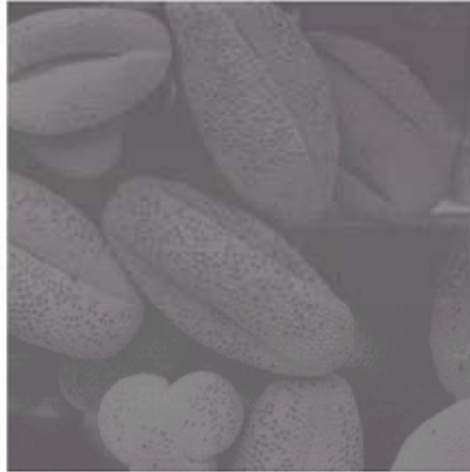
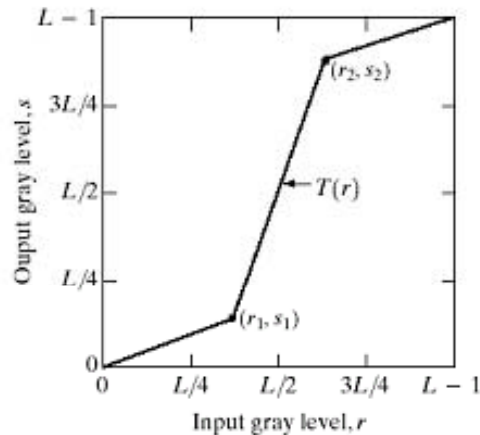
- (a) image has a washed-out appearance, it needs a compression of gray levels $\Rightarrow$ needs $\gamma > 1$
- (b) result after power-law transformation with $\gamma = 3.0$ (suitable)
- (c) transformation with $\gamma = 4.0$ (suitable)
- (d) transformation with $\gamma = 5.0$ (high contrast, the image has areas that are too dark, some detail is lost)



Piecewise-Linear Transformation Functions

- Advantage:
 - The form of piecewise functions can be arbitrarily complex
- Disadvantage:
 - Their specification requires considerably more user input

Contrast Stretching



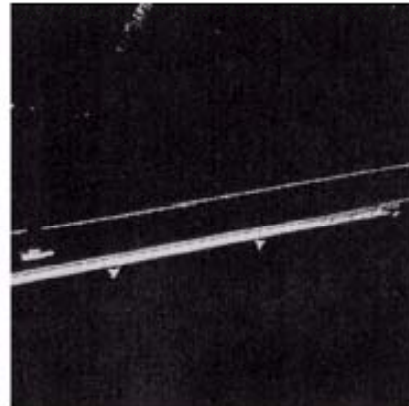
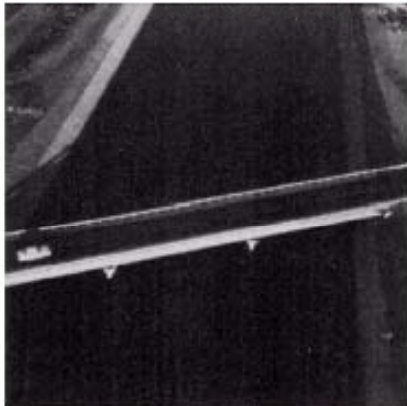
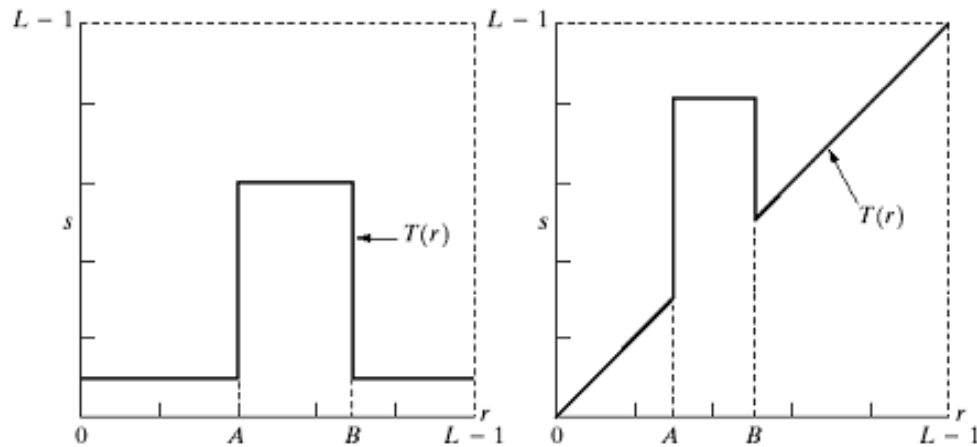
(a) Increase the dynamic range of the gray levels in the image

(b) a low-contrast image : result from poor illumination, lack of dynamic range in the imaging sensor, or even wrong setting of a lens aperture of image acquisition

(c) result of contrast stretching:
 $(r_1, s_1) = (r_{\min}, 0)$ and $(r_2, s_2) = (r_{\max}, L-1)$

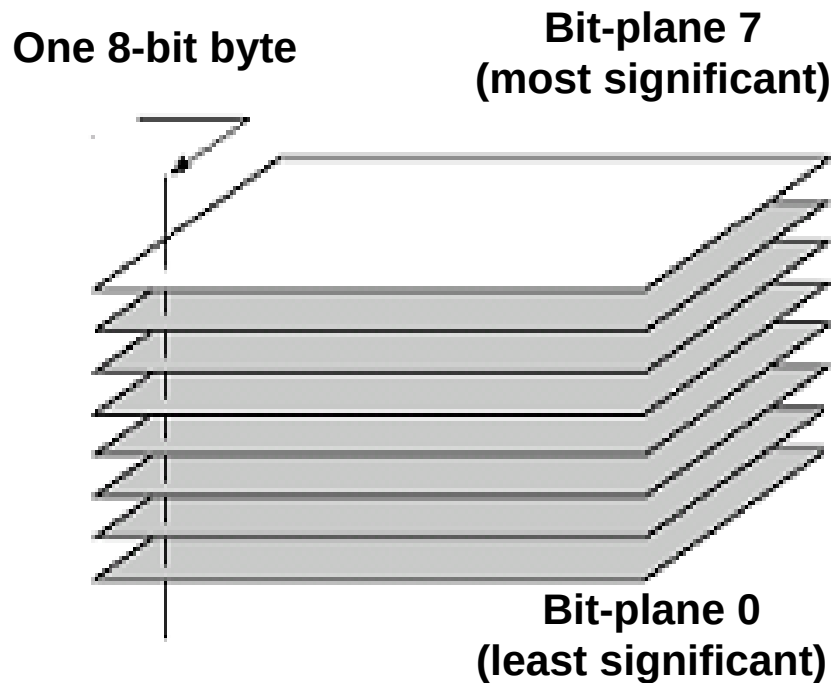
(d) result of thresholding

Gray-level slicing



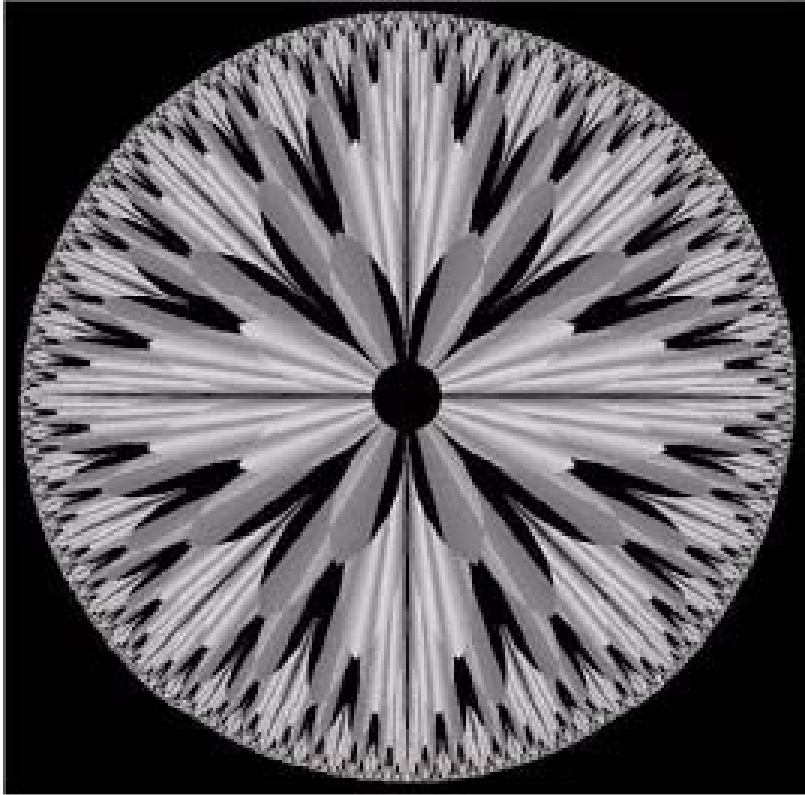
- Highlighting a specific range of gray levels in an image
 - Display a high value of all gray levels in the range of interest and a low value for all other gray levels
- (a) transformation highlights range $[A, B]$ of gray level and reduces all others to a constant level
- (b) transformation highlights range $[A, B]$ but preserves all other levels

Bit-plane slicing



- Highlighting the contribution made to total image appearance by specific bits
- Suppose each pixel is represented by 8 bits
- Higher-order bits contain the majority of the visually significant data
- Useful for analyzing the relative importance played by each bit of the image

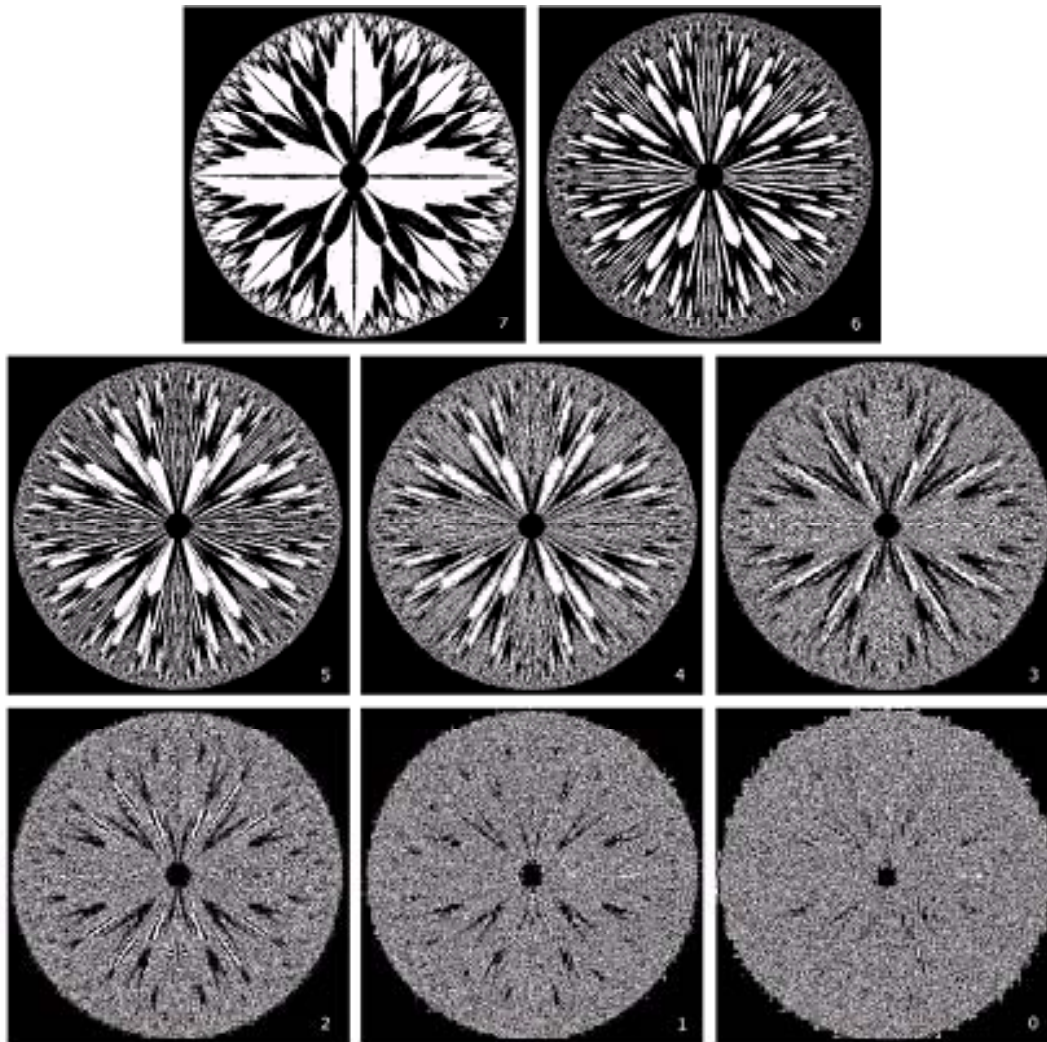
Example



An 8-bit fractal image

- The (binary) image for bit-plane 7 can be obtained by processing the input image with a thresholding gray-level transformation.
 - Map all levels between 0 and 127 to 0
 - Map all levels between 129 and 255 to 255

8 bit planes



Bit-plane 7		Bit-plane 6	
Bit-plane 5	Bit-plane 4	Bit-plane 3	
Bit-plane 2	Bit-plane 1	Bit-plane 0	

Histogram Processing

- Histogram of a digital image with gray levels in the range $[0, L-1]$ is a discrete function

$$h(r_k) = n_k$$

- Where
 - r_k : the k^{th} gray level
 - n_k : the number of pixels in the image having gray level r_k
 - $h(r_k)$: histogram of a digital image with gray levels r_k

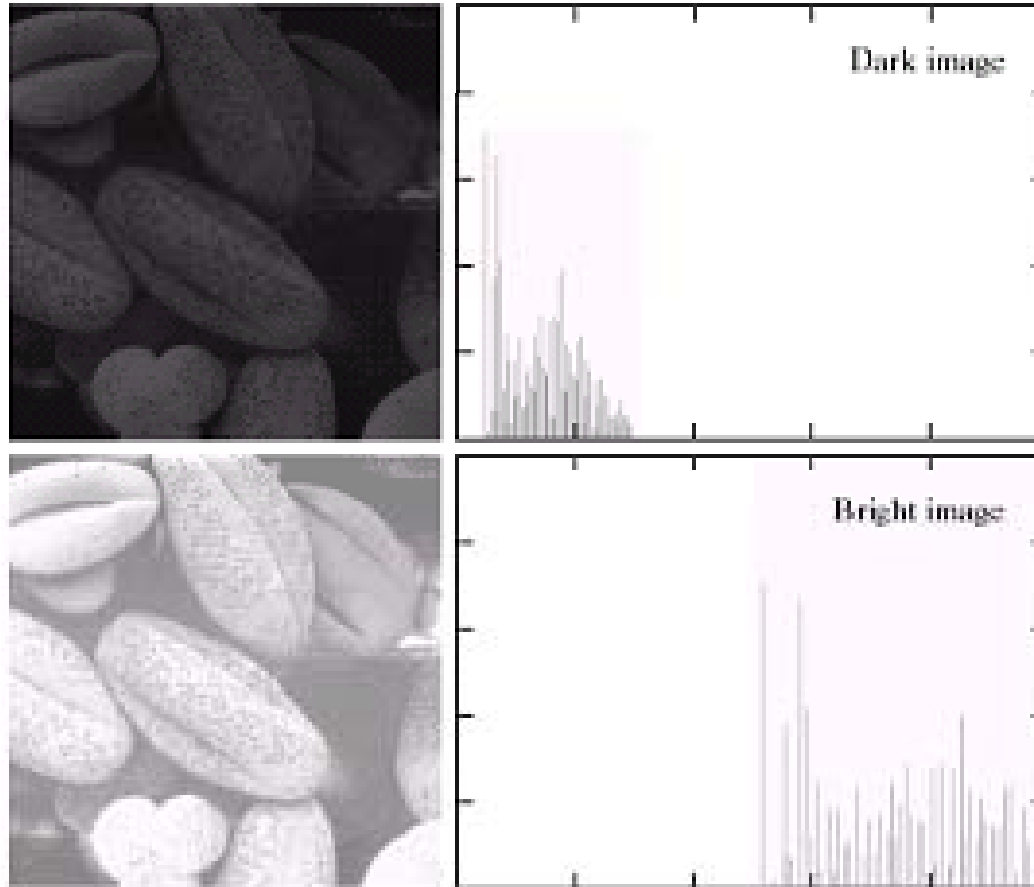
Normalized Histogram

- dividing each of histogram at gray level r_k by the total number of pixels in the image, n

$$p(r_k) = n_k / n$$

- For $k = 0, 1, \dots, L-1$
- $p(r_k)$ gives an estimate of the probability of occurrence of gray level r_k
- The sum of all components of a normalized histogram is equal to 1

Example



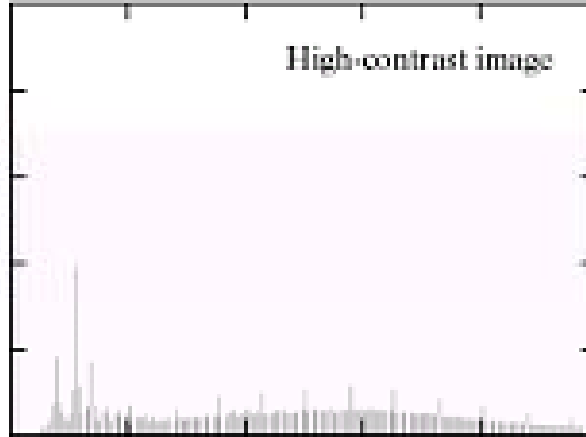
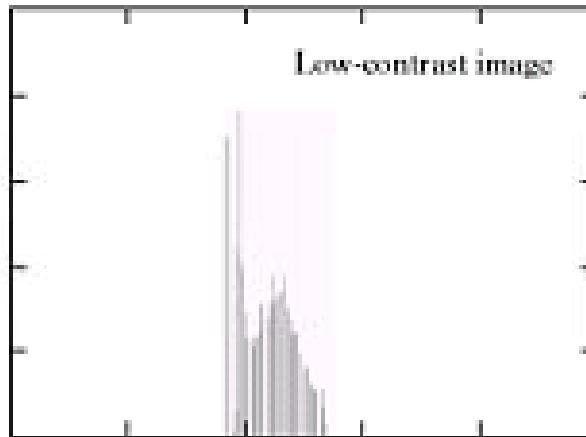
Dark image

Components of histogram are concentrated on the low side of the gray scale.

Bright image

Components of histogram are concentrated on the high side of the gray scale.

Example



Low-contrast image

histogram is narrow
and centered toward
the middle of the
gray scale

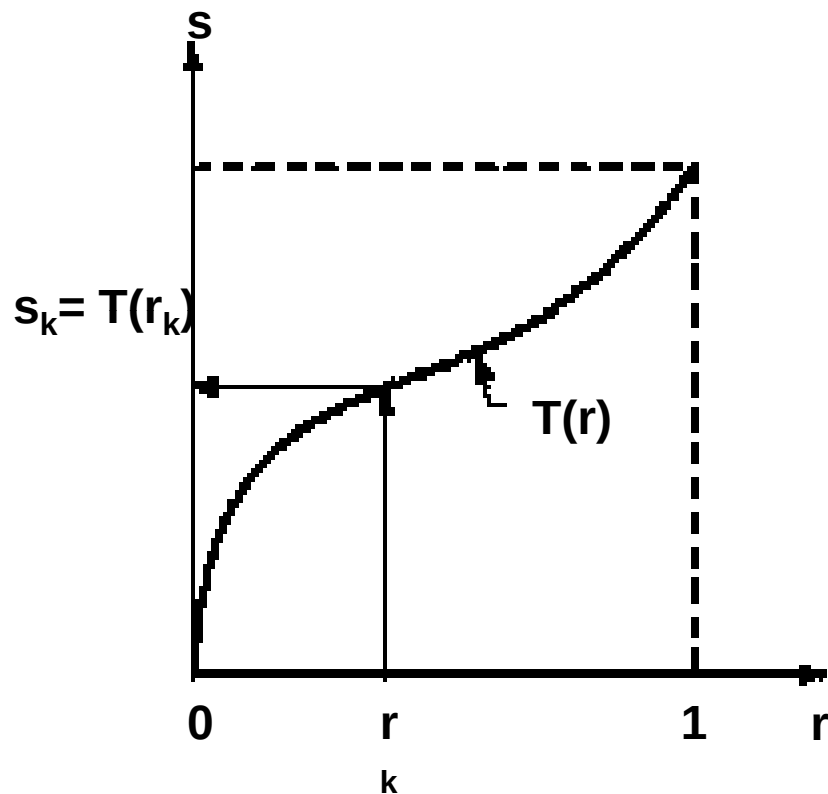
High-contrast image

histogram covers broad
range of the gray scale
and the distribution of
pixels is not too far from
uniform, with very few
vertical lines being much
higher than the others

Histogram Equalization

- As the low-contrast image's histogram is narrow and centered toward the middle of the gray scale, if we distribute the histogram to a wider range the quality of the image will be improved.
- We can do it by adjusting the probability density function of the original histogram of the image so that the probability spread equally

Histogram transformation



$s = T(r)$, where $0 \leq r \leq 1$ and satisfies

(a). $T(r)$ is single-valued and monotonically increasing in the interval $0 \leq r \leq 1$

(b). $0 \leq T(r) \leq 1$ for $0 \leq r \leq 1$

If (a) is satisfied, then the inverse transformation from s back to r is

$$r = T^{-1}(s) ; 0 \leq s \leq 1$$

How to determine $T(r)$ based on histogram

Probability Theory for Histogram processing

- Random variable x , the probability of x choosing a value less or equal to a is denoted by

$$F(a) = P(x \leq a), -\infty < a < \infty$$

- Function F is called the cumulative probability distribution function or simply the Cumulative Distribution Function (CDF), or simply distribution function.
- The properties of CDF:
 1. $F(-\infty) = 0$
 2. $F(\infty) = 1$
 3. $0 \leq F(x) \leq 1$
 4. $F(x_1) \leq F(x_2)$ if $x_1 < x_2$
 5. $P(x_1 < x \leq x_2) = F(x_2) - F(x_1)$
 6. $F(x+) = F(x)$

Random Variables

The Probability Density Function (PDF or shortly called density function) of random variable x is defined as the derivative of the CDF:

$$p(x) = \frac{dF(x)}{dx}$$

Properties of CDF

1. $p(x) \geq 0$ for all x
2. $\int_{-\infty}^{\infty} p(x) dx = 1$
3. $F(x) = \int_{-\infty}^x p(\alpha) d\alpha$, where α is a dummy variable
4. $P(x_1 < x \leq x_2) = \int_{x_1}^{x_2} p(x) dx$.

Random Variables

- If a random variable x is transformed by a monotonic transformation function to produce a new random variable y ,
- the probability density function of y can be obtained from knowledge of $T(x)$ and the probability density function of x , as follows:

$$p_y(y) = p_x(x) \left| \frac{dx}{dy} \right|$$

CDF Transformation Function

- A transformation function is a cumulative distribution function (CDF) of random variable r :

$$s = T(r) = F(r) = \int_0^r p(w)dw$$

- (a). $T(r)$ is single-valued and monotonically increasing
- (b). $0 \leq T(r) \leq 1$ for $0 \leq r \leq 1$

If (a) is satisfied, then the inverse transformation from s back to r is $r = T^{-1}(s)$;
 $0 \leq s \leq 1$

Discrete Random Variables

- Random variable x is discrete if the probability of choosing a sequence of numbers is equal to one.
- Suppose the sequence of numbers are x_i , $i=1, 2, \dots, N$, $x_i < x_{i+1}$.

Let $p(x_i)$ denote the probability of choosing x_i , $i=1, 2, \dots, N$.

- Then PDF

$$p(x) = \sum_{i=1}^N p(x_i) \delta(x - x_i), -\infty < x < \infty$$

- CDF

$$F(a) = \sum_{x_i \leq a} p(x_i)$$

Histogram Equilization

- The probability of occurrence of gray level in an image is approximated by histogram, i.e. the discrete version of transformation

$$p(r_k) \approx \frac{n_k}{n}, \quad \text{where } k = 0, 1, \dots, L-1$$

$$s_k = T(r_k) = F(r_k) = \sum_{j=0}^k p(r_j) = \sum_{j=0}^k \frac{n_j}{n} \quad \text{where } k = 0, \dots, L-1$$

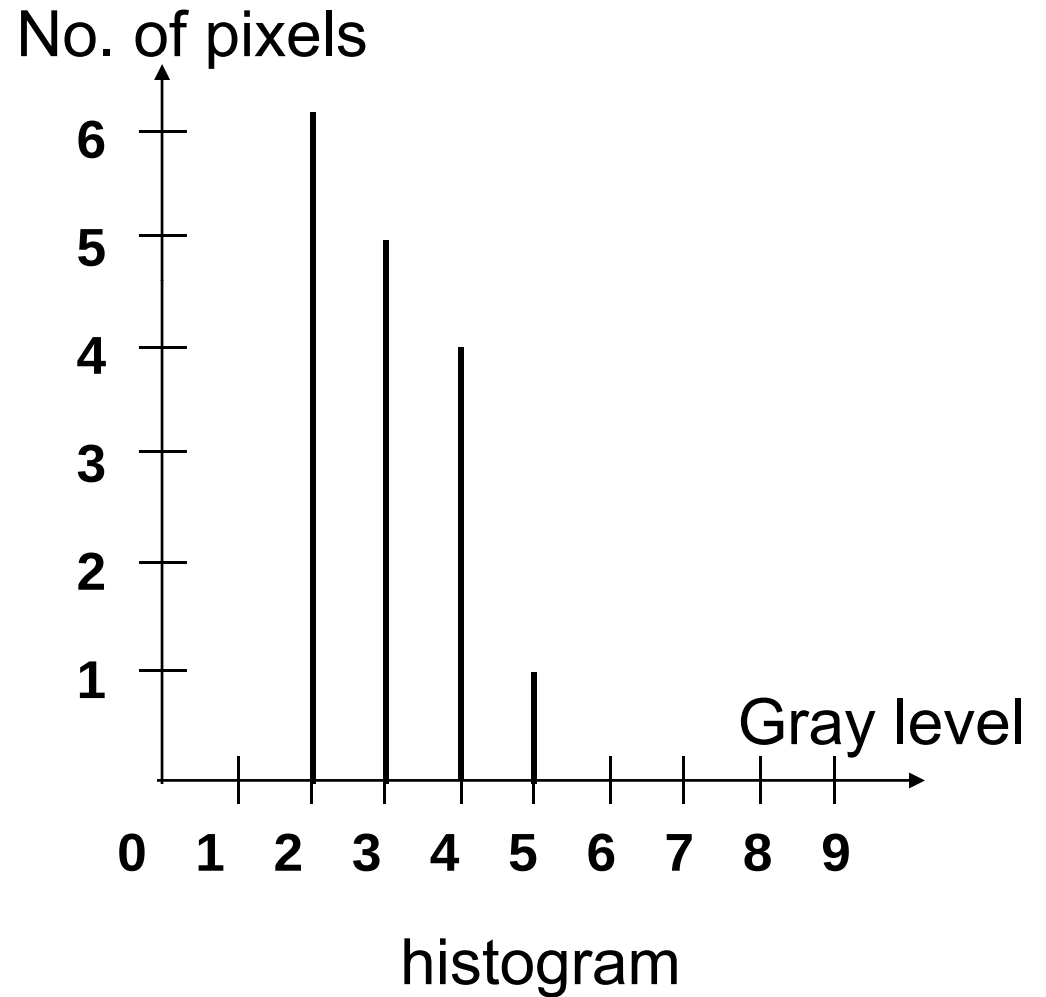
$$k \rightarrow \left(\sum_{j=0}^k \frac{n_j}{n} \right) \times (L-1)$$

Example

2	3	3	2
4	2	4	3
3	2	3	5
2	4	2	4

4x4 image

Gray scale = [0,9]

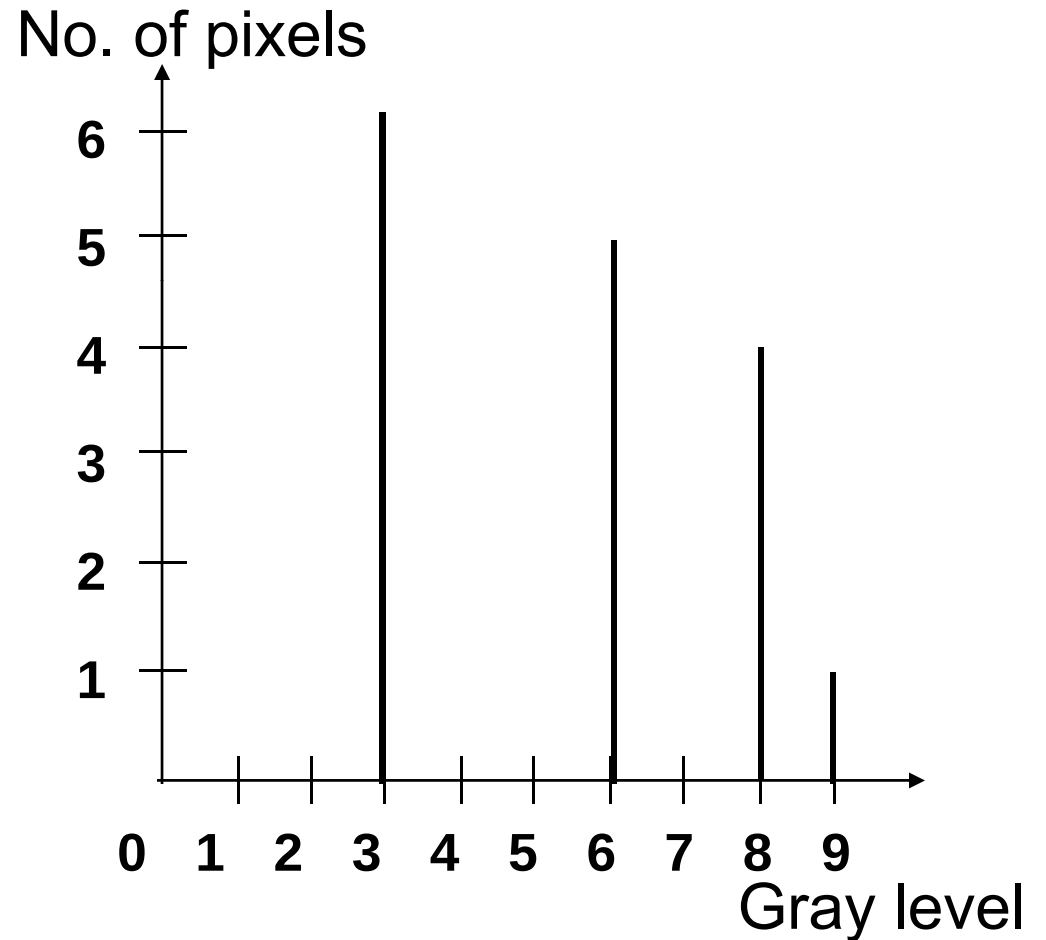


Gray Level(j)	0	1	2	3	4	5	6	7	8	9
No. of pixels	0	0	6	5	4	1	0	0	0	0
$\sum_{j=0}^k n_j$	0	0	6	11	15	16	16	16	16	16
$s = \sum_{j=0}^k \frac{n_j}{n}$	0	0	$\frac{6}{16}$	$\frac{11}{16}$	$\frac{15}{16}$	$\frac{16}{16}$	$\frac{16}{16}$	$\frac{16}{16}$	$\frac{16}{16}$	$\frac{16}{16}$
s x 9	0	0	$\frac{3.3}{\approx 3}$	$\frac{6.1}{\approx 6}$	$\frac{8.4}{\approx 8}$	9	9	9	9	9

Example

3	6	6	3
8	3	8	6
6	3	6	9
3	8	3	8

Output image
Gray scale = [0,9]



Histogram equalization

Histogram equalization

before



after

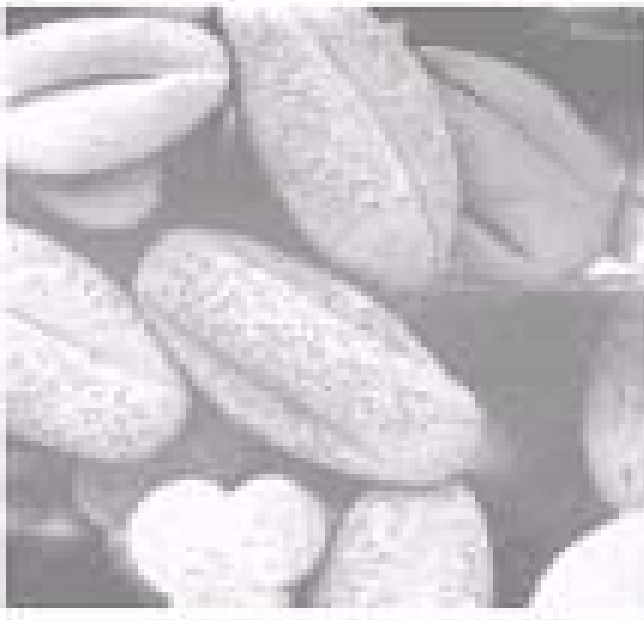
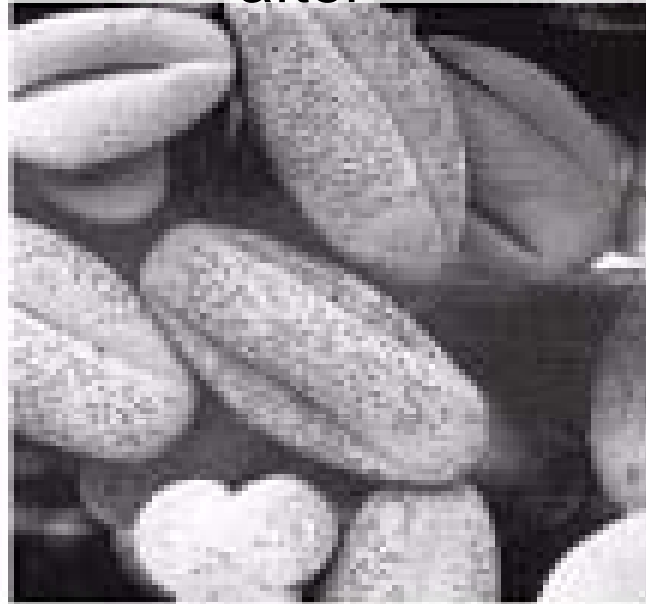
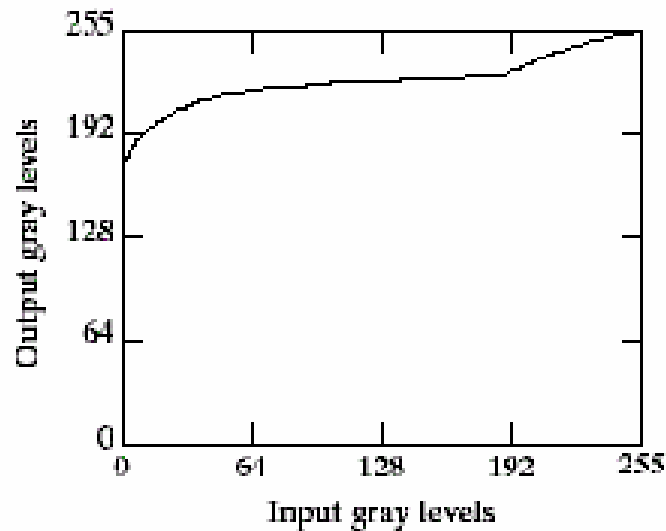
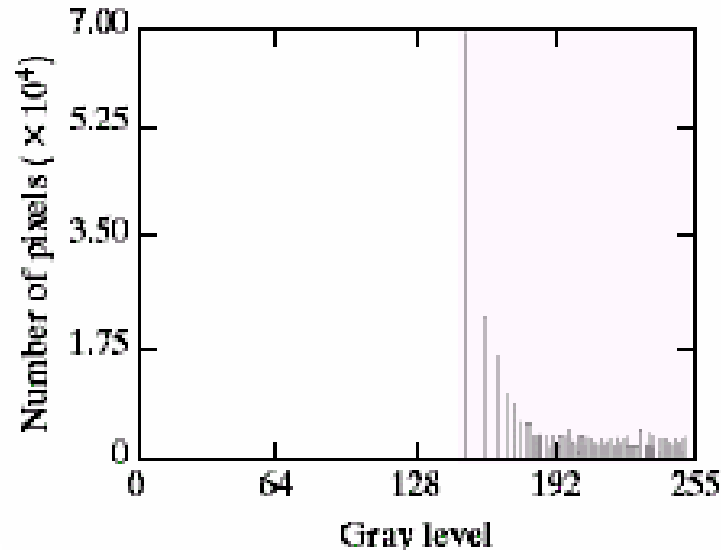


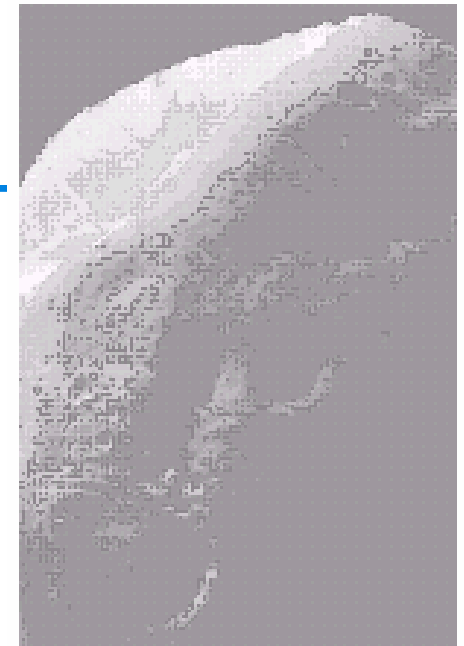
Image Equalization



Transformation function
for histogram equalization



Histogram of the result image



Result image
after histogram
equalization

The histogram equalization doesn't make the result image look better than the original image. Consider the histogram of the result image, the net effect of this method is to map a very narrow interval of dark pixels into the upper end of the gray scale of the output image. As a consequence, the output image is light and has a washed-out appearance.

Note

- Histogram equalization distributes the gray level to reach the maximum gray level (white) because the cumulative distribution function equals 1 when $0 \leq r \leq L-1$
- If the cumulative numbers of gray levels are slightly different, they will be mapped to little different or same gray levels as we may have to approximate the processed gray level of the output image to integer number
- Thus the discrete transformation function can't guarantee the one to one mapping relationship

Histogram Specification

- Histogram equalization has a disadvantage which is that it can generate only one type of output image.
- With Histogram Specification, we can specify the shape of the histogram that we wish the output image to have.
- It doesn't have to be a uniform histogram

Consider the continuous domain

Let $p_r(r)$ denote continuous probability density function of gray-level of input image, r

Let $p_z(z)$ denote desired (specified) continuous probability density function of gray-level of output image, z

Let s be a random variable with the property

$$s = T(r) = \int_0^r p_r(w) dw \quad \Rightarrow \quad \text{Histogram equalization}$$

Where w is a dummy variable of integration

Next, we define a random variable z with the property

$$g(z) = \int_0^z p_z(t) dt = s \quad \Rightarrow \quad \text{Histogram equalization}$$

thus

$$s = T(r) = G(z)$$

Therefore, z must satisfy the condition

$$z = G^{-1}(s) = G^{-1}[T(r)]$$

Assume G^{-1} exists and satisfies the condition (a) and (b)

We can map an input gray level r to output gray level z

Procedure Conclusion

1. Obtain the transformation function $T(r)$ by calculating the histogram equalization of the input image

$$s = T(r) = \int_0^r p_r(w)dw$$

2. Obtain the transformation function $G(z)$ by calculating histogram equalization of the desired density function

$$G(z) = \int_0^z p_z(t)dt = s$$

Procedure Conclusion

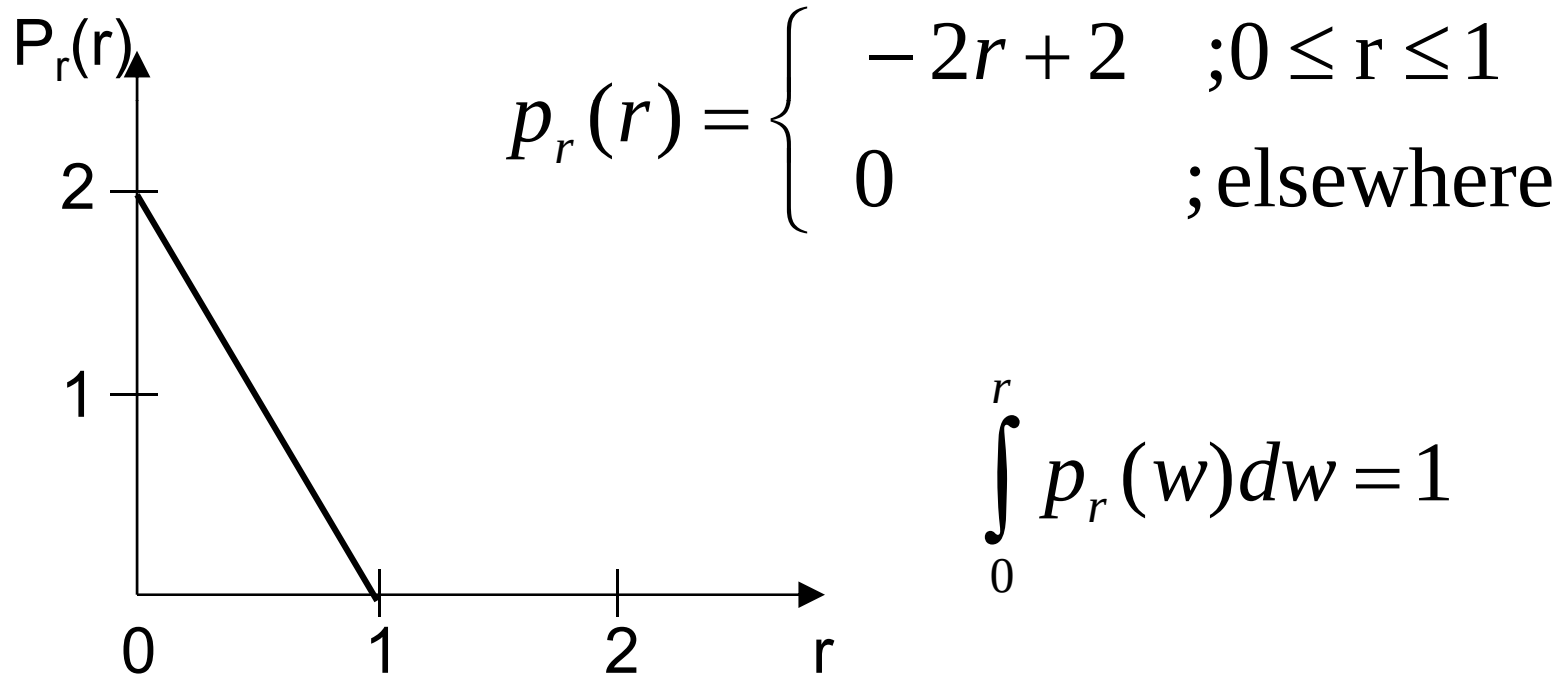
3. Obtain the inversed transformation function G^{-1}

$$z = G^{-1}(s) = G^{-1}[T(r)]$$

4. Obtain the output image by applying the processed gray-level from the inversed transformation function to all the pixels in the input image

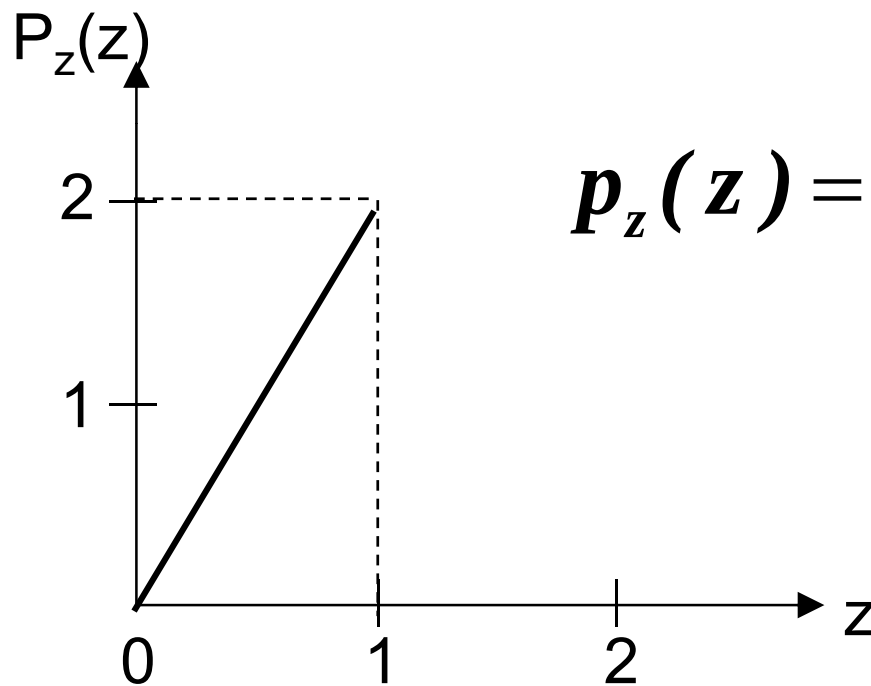
Example

Assume an image has a gray level probability density function $p_r(r)$ as shown.



Example

We would like to apply the histogram specification with the desired probability density function $p_z(z)$ as shown.

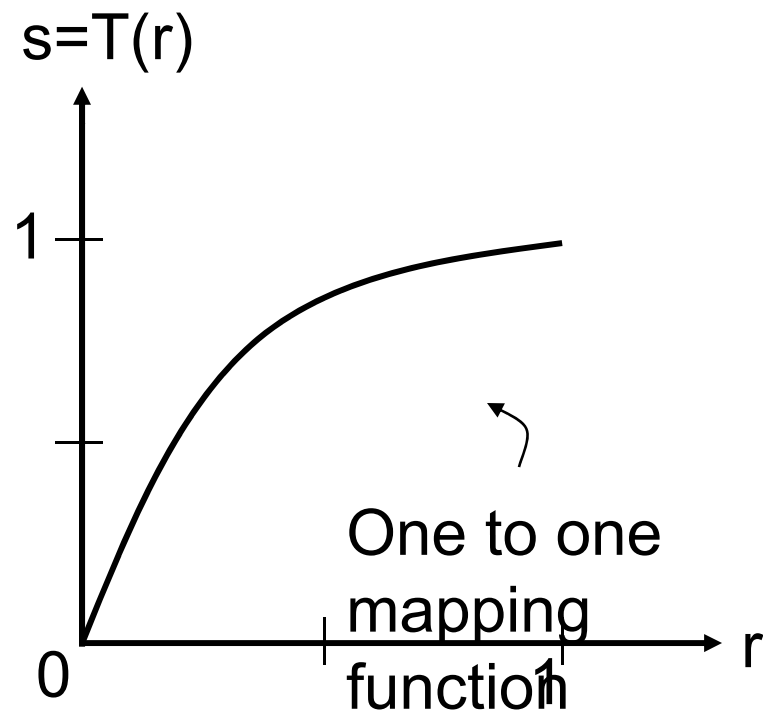


$$p_z(z) = \begin{cases} 2z & ; 0 \leq z \leq 1 \\ 0 & ; \text{elsewhere} \end{cases}$$

$$\int_0^z p_z(w) dw = 1$$

Step 1:

Obtain the transformation function $T(r)$



$$s = T(r) = \int_0^r p_r(w) dw$$

$$= \int_0^r (-2w + 2) dw$$

$$= -w^2 + 2w \Big|_0^r$$

$$= -r^2 + 2r$$

Step 2:

Obtain the transformation function $G(z)$

$$G(z) = \int_0^z (2w)dw = w^2 \Big|_0^z = z^2$$

Step 3:

Obtain the inversed transformation function G^{-1}

$$G(z) = T(r)$$

$$z^2 = -r^2 + 2r$$

$$z = \sqrt{2r - r^2}$$

We can guarantee that $0 \leq z \leq 1$ when $0 \leq r \leq 1$

Discrete formulation

$$\begin{aligned} \mathbf{s}_k &= \mathbf{T}(\mathbf{r}_k) = \sum_{j=0}^k \mathbf{p}_r(\mathbf{r}_j) \\ &= \sum_{j=0}^k \frac{\mathbf{n}_j}{\mathbf{n}} \quad k = 0, 1, 2, \dots, L-1 \end{aligned}$$

$$\mathbf{G}(\mathbf{z}_k) = \sum_{i=0}^k \mathbf{p}_z(\mathbf{z}_i) = \mathbf{s}_k \quad k = 0, 1, 2, \dots, L-1$$

$$\begin{aligned} \mathbf{z}_k &= \mathbf{G}^{-1}[\mathbf{T}(\mathbf{r}_k)] \\ &= \mathbf{G}^{-1}[\mathbf{s}_k] \quad k = 0, 1, 2, \dots, L-1 \end{aligned}$$

Example



Image of Mars moon

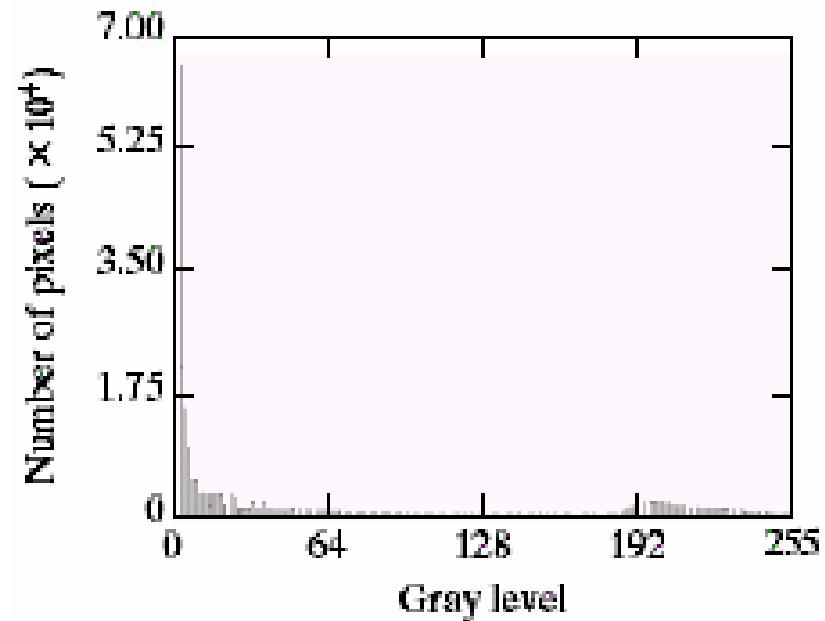
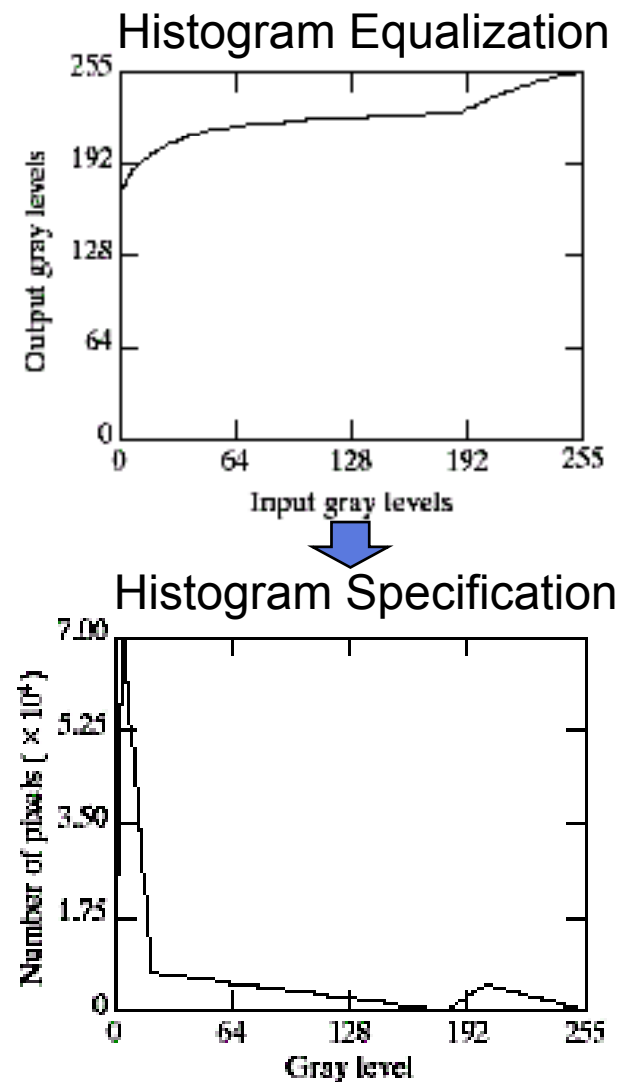


Image is dominated by large, dark areas, resulting in a histogram characterized by a large concentration of pixels in pixels in the dark end of the gray scale

Solve the problem

Since the problem with the transformation function of the histogram equalization was caused by a large concentration of pixels in the original image with levels near 0

a reasonable approach is to modify the histogram of that image so that it does not have this property

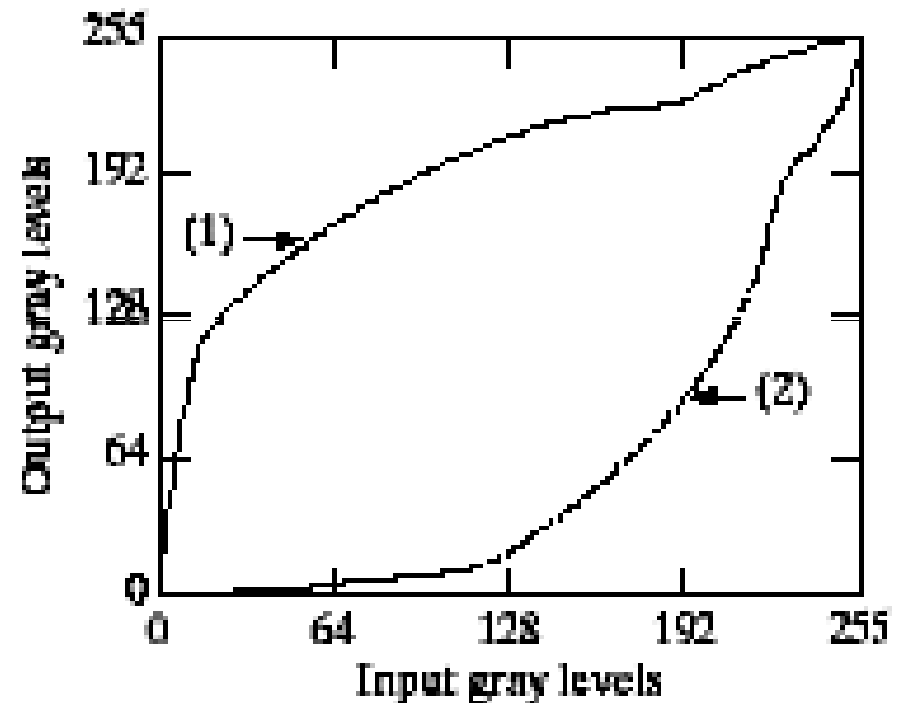


Histogram Specification

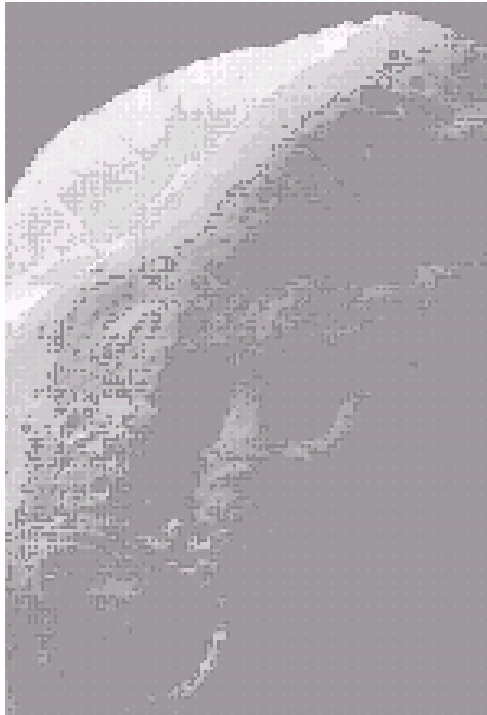
(1) the transformation function $G(z)$ obtained from

$$G(z_k) = \sum_{i=0}^k p_z(z_i) = s_k$$
$$k = 0, 1, 2, \dots, L-1$$

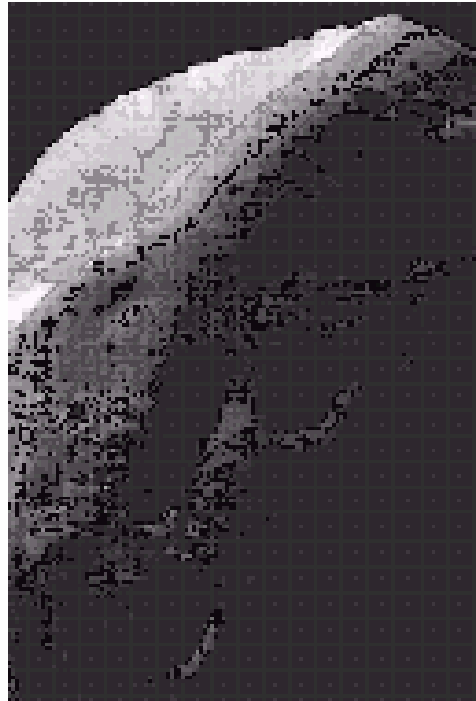
(2) the inverse transformation $G^{-1}(s)$



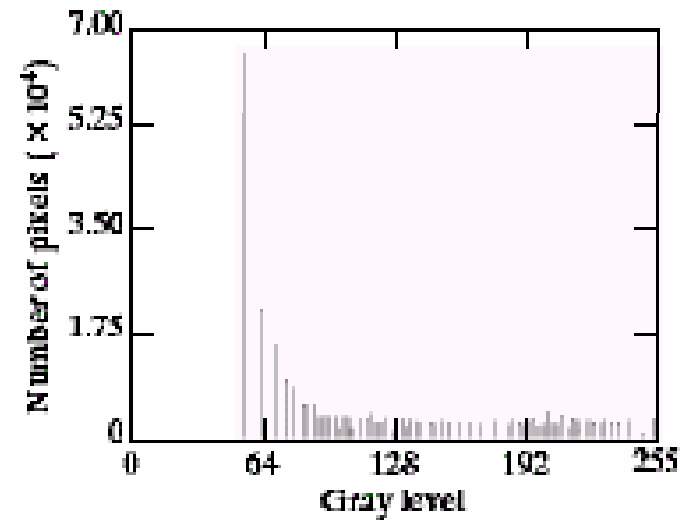
Result image and its histogram



Original image



After applied
the histogram
equalization



The output image's histogram

Notice that the output histogram's low end has shifted right toward the lighter region of the gray scale as desired.